Deep Reinforcement Learning (DRL) for Optimal Reservoir Control

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INTRODUCTION

Limited freshwater resources continue to become stressed due to ever-increasing demands from rapid urbanization, expanding agriculture and climate change. This scarcity exacerbates instability and conflicts both within the US and globally. Moreover, complex water rights agreements complicate existing water management strategies designed for historical climate regimes that have changed and are expected to continue to change. Existing reservoir management strategies can be complemented by efficient data-driven, machine learning approaches to optimize water and hydropower resources that account for dynamically evolving objectives and constraints and an unsteady climate. We propose a reservoir water release control framework utilizing deep reinforcement learning (DRL), consisting of a decision-making "agent" (a neural network-based algorithm) that interacts with a virtual physical environment to provide optimal dam management policies. This framework caters to multiple objectives and is flexible to changing forecasts.

Our DRL model aims to serve as an advanced tool that could assist dam operators in making more informed and efficient decisions. By providing real-time recommendations, the model seeks to enhance operational decision-making while preserving the expertise and judgment of human operators. We are currently using the United States Bureau of Reclamation (USBR) data for the Navajo reservoir **(Figure 1)** to test the proposed framework. Although relatively small, the Navajo is crucial to New Mexico and Colorado owing to its role in flood control, hydropower generation, irrigation and ecosystem protection.

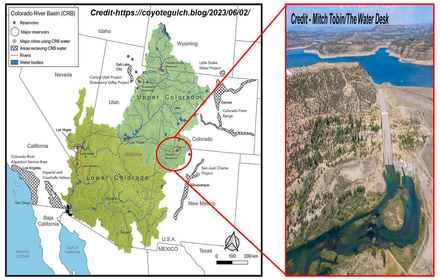

Figure 1. Navajo Reservoir

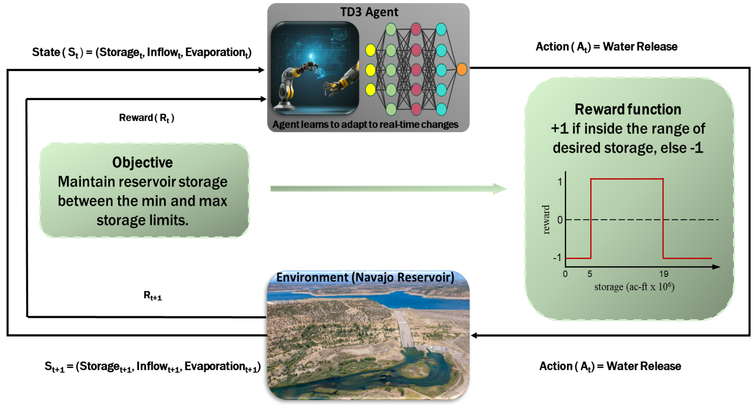

Figure 2. Proposed DRL Framework

METHODOLOGY

Our DRL framework (**Figure 2**) consists of a Twin Delayed Deep Deterministic (TD3) (**Figure 4**) agent [*https://spinningup.openai.com/en/latest/algorithms/td3.html*] that receives a set of state or observation variables (inflow, evaporation, storage) from the environment (reservoir) and, in turn, takes action(adjusting water release). As a consequence of this action, the agent receives a reward and a set of new environment states (adjusted reservoir levels, evaporation, inflow). The agent-environment interaction continues and helps the agent learn an optimal behavior. The next step is to use machine learning-based models such as the Long Short-Term Memory (LSTM) networks (**Figure 3**) to provide the inflow (streamflow) and evaporation values based on the input meteorological features pertinent to the local geography. We conducted two experiments to demonstrate the efficacy of the DRL agent.:

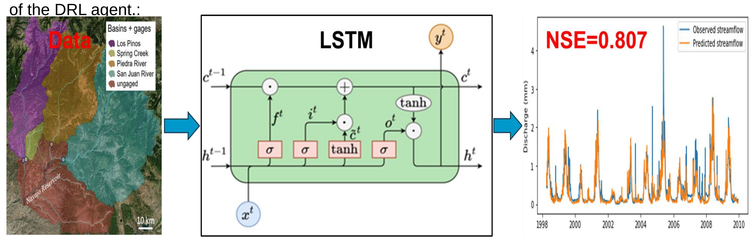

Figure 3. LSTM model for streamflow (inflow) prediction

1. LEARNING HUMAN BEHAVIORAL PATTERN

In the first experiment, we used the DRL agent (experimented with different agents) to imitate the human (dam operator) decision-making. For this experiment, the DRL agent takes inflow, evaporation, and storage values as inputs and outputs the volume of water released**.**

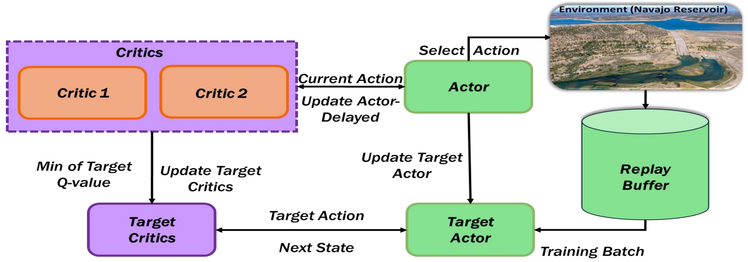

Figure 4. TD3 Agent Architecture

2. OPTIMIZING RESERVOIR OPERATIONS

In the second experiment, we used the DRL agent to optimize the decision-making process involved in reservoir operations. Using the data provided by the USBR collaborators, the DRL agent receives state variables, namely, storage, inflow, and evaporation, as inputs and takes action to release water. In this experiment, the DRL agent does not explicitly try to match the release made by the human operator; instead, the agent tries to learn optimal behavior (e.g., store more water for the dry months) for set objectives while abiding by the operational and design constraints under changing meteorological conditions**. Figure 6** demonstrates the corresponding results

RESULTS

1. LEARNING HUMAN BEHAVIORAL PATTERN

The DRL agent registers a mean squared error (MSE) < 0.01 while imitating human behavioral patterns when releasing water from the reservoir. Despite the complex intricacies of human decision-making involved when releasing water from the reservoir, **Figure 5** shows that the DRL agent's predicted release traces the actual human operator-made release quite well.

Type of DRL Agent (Algorithm)	Mean Squared Error (MSE)
PPO	0.034231
SAC	0.015
TD3	0.00953

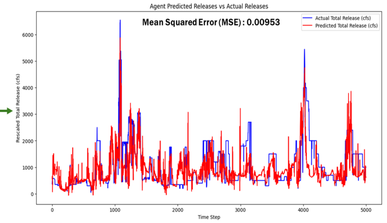

Figure 5. Performance when learning human behavioral pattern

2. OPTIMIZING RESERVOIR OPERATIONS

The objective of the DRL agent was to release water from the reservoir while maintaining the water volume within the min-max storage limits, which were 500000 AF and 1900000 AF, respectively. We used the most basic binary reward function to guide the training of the DRL agent. The agent gets a reward of '1' for staying within the storage limits, while it gets penalized by '-1' if the water volume exceeds the min-max storage limits. **Figure 6(a)** shows the predicted and actual storage values. The plot shows that even the most basic binary reward function helps the agent learn a release pattern (**Figure 6(b)**) while maintaining the water volume within the min-max storage limits.

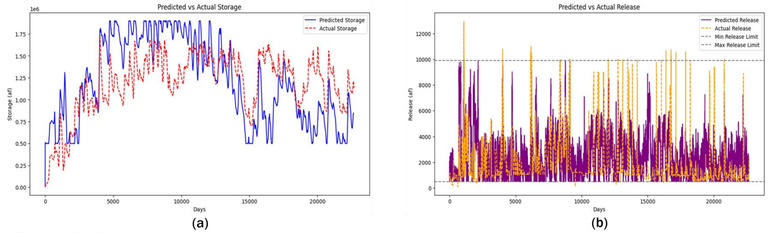

Figure 6. Performance when optimizing reservoir operations

DISCUSSION & FUTURE WORK

As evident from our first experiment, the DRL agent is efficient when attempting to mimic the thought process of an actual human operator. The second experiment is a steppingstone towards the multi-objective optimization process. The plots in **Figure 6** show that the DRL agent maintains the storage levels within the desired min-max limits while releasing water from the reservoir. Another noteworthy point is that the basic binary reward function entirely guides the DRL agent's current learning. With the more complicated and comprehensive reward functions we are developing, we aim to improve performance while addressing multiple objectives, dynamically evolving climate conditions, and future water demands.